

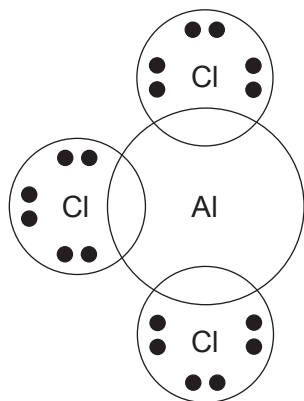
SECTION A*Answer all questions in the spaces provided.*

1. Complete the electronic structure of the Co^{2+} ion below. [1]

$1s^2 2s^2 2p^6 3s^2 3p^6$

2. The compound $\text{AlCl}_3 \cdot \text{NH}_3$ contains both covalent and coordinate bonds. [2]

Complete the dot and cross diagram for this compound.



3. Write the expression for the ionic product of water, K_w . [1]

4. The standard hydrogen electrode and hydrogen fuel cells both use the same metal as an electrode. Name the metal and explain why this metal is suitable. [1]

.....



5. When CCl_4 is added to water two separate layers are formed and there is no reaction, whilst SiCl_4 reacts violently with water.

Give the reason for this difference.

[1]

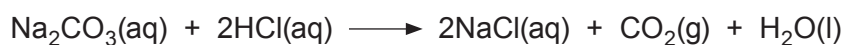
.....

.....

.....

6. The entropy change of the reaction shown below is positive. Give a reason for this.

[1]



.....

.....

7. Carbon monoxide can be used to extract iron by the reduction of Fe_3O_4 .

(a) Write an equation for this reaction.

[1]

.....

(b) Give a reason why carbon monoxide acts as a reducing agent while the equivalent oxide of lead does not.

[1]

.....

.....

8. Draw the **structure** of the copper-containing ion formed when concentrated hydrochloric acid is added to a solution containing $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ ions.

[1]

